macOS Computer Literacy
Setup and Everyday Reference

UCC Physics Society

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1 Install VS Code or VSCodium

Choose one editor. For VS Code, download the disk image and move `Visual Studio Code.app` to Applications using the official macOS setup guide. For VSCodium, follow the official installation options; the Homebrew command is available if Homebrew is already installed. Neither Homebrew nor an administrator account is required for this workshop.

2 Install the shared tools

- Install Python from the official Python macOS downloads. Use `python3`; do not modify Apple-managed system files.
- Run `git -version`. If macOS offers to install Command Line Tools, follow Apple's prompt; full Xcode is not required just to use Git.
- For local LaTeX, install MacTeX. It is a large download; Overleaf is a browser alternative.

These vendor pages provide current installation choices: Git for macOS, Homebrew, and TeX Live. Homebrew is optional, not a prerequisite. After installing MacTeX, reopen Terminal before checking `pdflatex -version` and `latexmk -v`.

```
python3 --version
git --version
pwd
```

3 Terminal, projects, and paths

Open Terminal from Spotlight. Recent macOS versions use **zsh** by default. The terminal runs a shell to navigate folders and start programs. Relative paths begin from the current working directory. Finder's `Command+Shift+G` opens a path; `Command+Shift+.` toggles hidden files. Treat filenames as case-sensitive for portable work.

```
pendulum-lab/
  README.md
  requirements.txt
  data/
    raw_measurements.csv
  analysis/
    analysis.py
    analysis.ipynb
  figures/
  report/
    report.tex
```

Task	Terminal command
Current directory	`pwd`
List files, including hidden	`ls -la`
Move to the parent directory	`cd ..`
Go to your home folder	`cd ~`
Enter a path with spaces	`cd "My Project"`
Copy / move a file	`cp source.csv copy.csv` / `mv copy.csv data.csv`
Find CSV files	`find . -name '*.csv'`
Search file text	`grep -n 'time' notes.txt`
Open the current folder in Finder	`open .`
Open the folder in the editor	`code .` or `codium .`

Inspect paths before copying or moving files: an existing destination may be overwritten. Prefer Finder's Trash when removing files. If the editor command is unavailable, use File > Open Folder or install its shell command from the editor's Command Palette. See Apple's Terminal User Guide for opening Terminal, working with windows, and using commands.

4 Python and project environments

Create one virtual environment in the project directory containing `requirements.txt`. Install project packages inside it:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m pip check
python -c "import sys; print(sys.executable)"
deactivate
```

The printed interpreter path should include `.venv`. A virtual environment keeps project packages separate; it is not a security sandbox, and it should not be committed or copied to another computer. See the Python virtual-environment documentation. Activation applies only to the current terminal.

5 Git and LaTeX

Git works locally; GitHub hosts online copies and collaboration. Saving a file is not the same as committing it, and a commit is not the same as pushing it. For the shared commands and safe workflow, see the Git and GitHub guide.

LaTeX source (`.tex`) is compiled into a PDF. Use Overleaf in a browser or install MacTeX for local builds. From the repository root, build this handout with:

```
latexmk -cd -pdf -interaction=nonstopmode -halt-on-error MacOS/MaxOS.TEX
```

The shared LaTeX guide covers report structure, equations, figures, and references; see also the Overleaf LaTeX introduction and LaTeX Project documentation.

6 Shortcuts and zsh configuration

- `Ctrl+C` interrupts a terminal command; `Tab` completes paths; `Up` recalls command history.
- `Command+C/V` copies and pastes in standard macOS applications.
- `Command+Space` opens Spotlight; `Command+Tab` switches applications.

zsh reads `~/.zshrc` for interactive shells. To add personal shortcuts, open the file in a text editor and append simple aliases:

```
alias ll='ls -lah'
alias croot='cd ~'
```

Save the file, then load the changes in the current terminal:

```
source ~/.zshrc
```

Back up and review your own file before editing it. Bash configuration is different; do not copy a Linux `.bashrc` wholesale. See the zsh startup-file documentation.

7 Troubleshooting

Python command missing: Use `python3` and check that the official installer completed.

Editor command missing: Use File > Open Folder or install the editor's shell command, then restart Terminal.

Python package installation is blocked: Check that the project environment is active.

File not found: Check `pwd`, filename case, and the relative path.

TeX command missing: Reopen Terminal after installing MacTeX; consult its official installation instructions before changing paths.

Application blocked: Verify its source and university policy; do not disable macOS security globally.

For Python, fitting, and data-analysis exercises, use the shared exercise notebook.